

to produce an audible sound indicating full level of water in the overhead tank, and switching off 230V AC supply to the pump motor given through switch S3.

If the pump motor is not switched off, pumping of water will continue and water will reach the overflow level. This makes water contact between CS1 and L1, making input pin 1 of IC1 high and output pin 16 low. Transistor T4 conducts through base resistor R20, providing 12V DC from the emitter to the collector. Relay RL1 energises, cutting off the operating 230V AC given to the pump motor.

Shift switches S2 and S3 to 'off' position when the tank is full.

When water in the overhead tank goes below the reserve level, sensor electrodes CS1 and L4 don't make contact through water, making input pin 4 of IC1 low and output pin 13 high. LED5 remains 'off' and the base of transistor T2 goes high, providing 12V DC from the collector to the emitter. This voltage operates piezobuzzer PZ1 through diode D6 to produce an audible alarm.

In the ground tank when the water level rises, water-sensing elec-

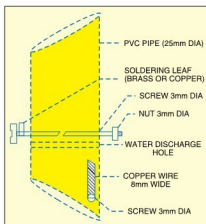


Fig. 6: Sensor assembly details

trodes CS2 and L7 through L5 are suspended in the ground tank to sense reserve, half and full levels of water in the ground tank. When water in the ground tank bridges CS2 with L7 through L5, respectively, pins 7 through 5 of IC1 in that sequence go high and the corresponding outputs at pins 10 through 12 go low. LED2 through LED4 glow to indicate different water levels in the ground tank.

As water is filled in the overhead tank, water level in the ground tank goes down. When water level in the

ground tank goes below the preset reserve level, sensor electrodes CS2 and L7 don't make contact through water, keeping input pin 10 of IC1 high and LED2 'off'. The base of transistor T1 also goes high, providing 12V DC from the collector to the emitter. This voltage is fed to piezobuzzer PZ1 through diode D5 to produce an audible alarm. The 12V DC from the emitter also energises relay RL2, turning the pump motor off by cutting off the operating 230V AC fed to it. Shift switches S2 and S3 to 'off' position.

Construction

An actual-size, single-side PCB for the water-tank-filling guard system for the overhead and ground tanks is shown in Fig. 2 and its components layout in Fig. 3. Assemble the components on the PCB. After ensuring that the component values are correct, enclose the PCB in a suitable cabinet. Connect sensors for overhead and ground tanks using flexible wires.

Fig. 4 shows the overhead tank sensor assembly, while Fig. 5 shows the ground tank sensor assembly. Refer Fig. 6 for sensor assembly details. ●

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KEY FACTS

Date: 17th to 19th February, 2011 Hours: 10:00 to 18:00
Venue: Hall No. 2, 3, 4, 5 and 6 at Pragati Maidan, New Delhi